

Decking, Underlayment and Slope

What every roof is made of, and the slope numbers that decide which covering is allowed on it.

Both pitched and flat roofs are made of the same two basic elements: decking and covering. The decking is the structural surface; the covering is what sheds the water. Between them sits the underlayment, and on a steep slope roof it is the underlayment that creates the actual waterproof barrier.

YOU HAVE MISSED THIS 2 TIMES

Steep slope roof coverings rely on _____ to create a waterproof barrier.

Underlayment.

This is worth understanding rather than memorizing. Shingles, shakes and tiles are not watertight on their own — they are overlapping pieces that shed most of the water and slow the rest down. The layer underneath is what actually keeps water out, which is why steep slope coverings need slope to work and flat roofs need a genuinely sealed membrane instead.

Naming the parts

Ridge	The horizontal line at the top of the roof, where two planes meet. A gable roof has one.
Hip	An external corner where two roof planes meet and slope away from each other.
Valley	The internal corner where two planes meet and drain into each other. It carries the most water on the roof.
Rake	The sloping edge of a gable end.
Eave	The lower edge, where the roof overhangs the wall.
Gable	The triangle of wall closed in by a two-sided sloping roof. A gable roof has a ridge.
Closed valley	A valley where the covering runs across the join, so the valley flashing is not visible.

The eave is the lowest part of the roof, the part that projects beyond the structure walls. Ridge at the top, eave at the bottom, rake down the sloping edge between them.

YOU HAVE MISSED THIS 3 TIMES

To protect the interior of a structure, a flat roof must have a/an:

Watertight covering.

That is the same idea from the other end. A steep roof sheds; a flat roof has no slope to shed with, so its covering itself must be watertight.

The slope numbers

The numbers here

Low slope, treated as a flat roof, below	2:12
Steep slope, above	4:12
Wood shake and shingle — minimum slope	4 in 12
Roof sheathing	5/8 in
Head lap	2 in

Slope is written as rise over run: 4:12 means four inches of rise for every twelve inches of horizontal travel. The steeper the roof, the faster water leaves and the more forgiving the covering can be. Below 2:12 water no longer runs off reliably, so the roof must be sealed rather than lapped.

Say it back

Cover the page. Answer each one out loud.

1. Name the two basic elements every roof is comprised of.
2. Ridge, hip, valley, rake, eave — say what each one is.
3. What must a flat roof have to protect the interior?
4. What does a closed valley mean?
5. What creates the waterproof barrier on a steep slope roof?
6. Why are shingles not watertight by themselves?
7. Give the low slope and steep slope numbers.
8. Minimum slope for wood shakes and shingles?
9. Standard roof sheathing thickness, and the head lap.